Chapter 2.

SCHOOL OF COMPUTING & DATA SCIENCE
The University of Hong Kong

Mobile App Development Lifecycle



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COMP3330 Interactive Mobile Application Design and Programming
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5 Major Phases

- Inception
- Design
- Development
- Stabilization / Testing
- Deployment

Inception (Strategy)

- To define and refine the idea of an app
- Some fundamental questions to ask:
  - Competitive Advantage
    - Are there similar apps? If so, how does this application differentiate from others?
  - Infrastructure Integration
    - For apps that will be distributed in an enterprise, what existing infrastructure will it integrate with or extend?
  - Value
    - What value does this app bring users? How will they use it?
  - Form / Mobility
    - How will this app work in a mobile form? How can I add value using mobile technologies such as location awareness, the camera, etc.?
- To help with designing the functionality of an app, it can be useful to define:
  - Actors: roles within an application and are often users
  - Use cases: typically, actions or intents

Inception (Analysis & Planning)

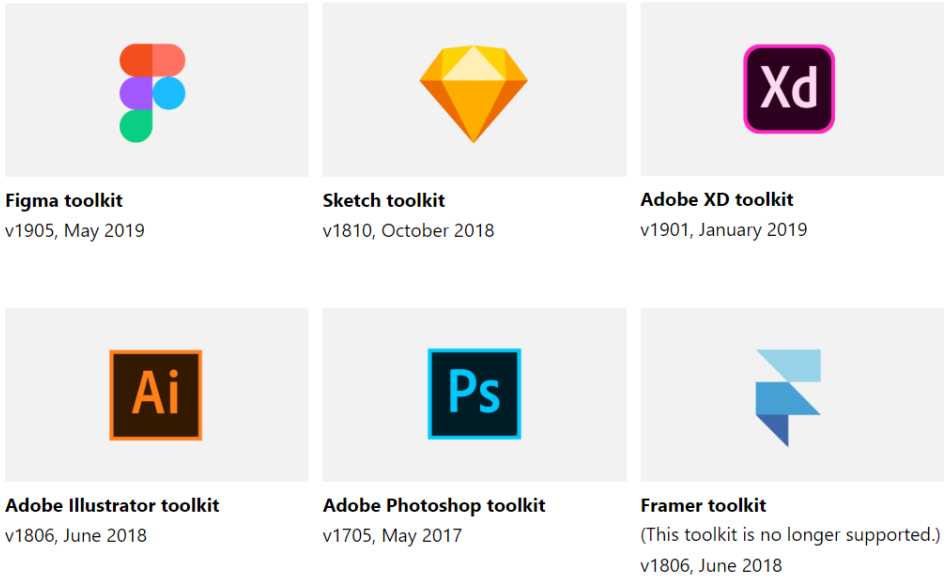
- Define clear objectives:
  - Each mobile application has to be designed for a specific purpose and marketing campaign.
    - What you want to achieve?
    - What's the expected outcome?
  - You cannot include a complete set of services or products into one mobile application.
  - Identify the exact scope of mobile application and then focus on how it can help you meet your objectives.
- Identify target audience and requirements:
  - Where do you think you can reach them?
  - What are their implicit needs?
  - What are the on-the-go features they might demand and how well can you satisfy their need?
  - The functionality and user interface must be in line with customer expectations.

Design – User Experience (UX) Design

- User Experience (UX) Design
  - UX is usually done via wireframes or mockups using any design toolkits. UX mockups allow the UX to be designed without having to worry about the actual UI design.



Wireframe



Design toolkits
(Reference: https://docs.microsoft.com/en-us/windows/uwp/design/downloads/)

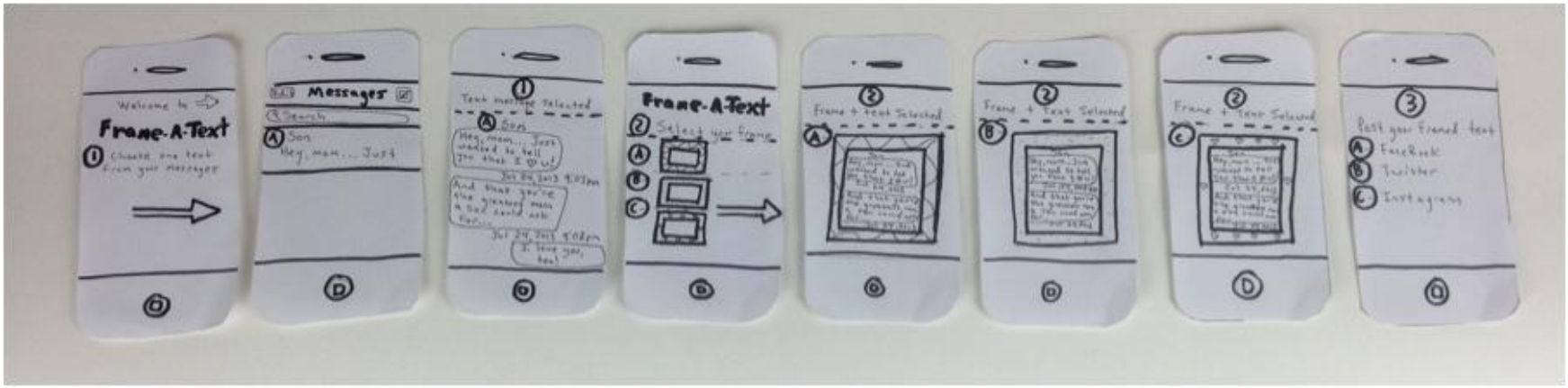
Paper-based Design Prototype

- Paper Prototyping is a technique that consists of creating hand drawings of user interfaces in order to enable them to be rapidly designed, simulated and tested.
- Paper prototypes are used:
  - to communicate ideas: between designers, developers, users and other stakeholders in the first stages of the user-centered design process.
  - as a usability testing technique: to observe the human interaction with user interfaces even before these interfaces are designed and developed.

Paper-based Design Prototype





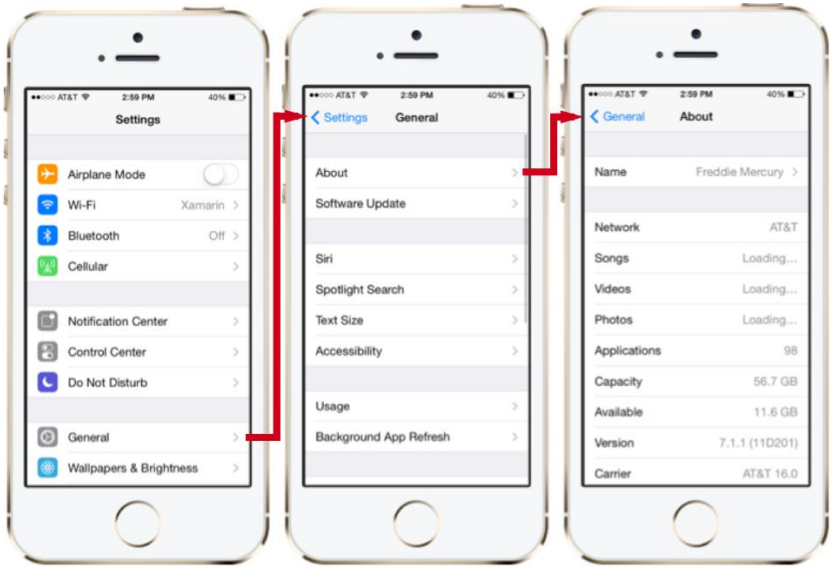


Design – User Experience (UX) Design

- User Experience (UX) Design
  - When creating UX mockups, you may need to consider the interface guidelines for the various platforms that the app will target. The app should "feel at home" on each platform.
  - Official design guidelines for each platform:
    - Apple - Human Interface Guidelines
      - https://developer.apple.com/design/human-interface-guidelines/ios/overview/themes/
    - Android – Design Guidelines
      - https://developer.android.com/design/index.html

Design – User Experience (UX) Design

- User Experience (UX) Design
  - The hardware itself also affects UX decisions.
  - Example 1: iOS devices have no physical back button, and therefore introduce the Navigation Controller metaphor.



  - Example 2: (Form factor issue) A tablet has bigger screen and so can display more information. Often what needs multiple screens on a phone is compressed into one for a tablet.

Design – User Interface (UI) Design

- User Interface (UI) Design
  - Once the UX is determined, the next step is to create the UI design.
  - Difference between UX Design and UI Design:
    - UX Design is typically just black and white mockups
    - UI Design is with colors, graphics, etc. Spending time on good UI design is important and generally, the most popular apps have a professional design.
  - As with UX, it's important to understand that different platforms are different, so a well-designed application may still look different on each platform.



Development

- The development phase usually starts very early.
- In fact, once an idea has become mature in the conceptual/inspiration phase, a working prototype (not just an animation) should be developed that validates functionality, assumptions, and helps to give an understanding of the scope of the work.

Stabilization / Testing

- Stabilization is the process of working out the bugs in your app. Not just from a functional standpoint, but also Usability and Performance.
- It's best to start stabilization very early within the development process so that corrections can occur before they become costly.
- Typically, an application should follow these patterns:
  - Prototype – The app is still in proof-of-concept phase and only core functionality, or specific parts of the application are working. Major bugs are present.
  - Alpha – Core functionality is generally code-complete (built, but not fully tested). Major bugs are still present, outlying functionality may still not be present.
  - Beta – Most functionality is now complete and has had at least light testing and bug fixing. Major known issues may still be present.
  - Release Candidate – All functionality is complete and tested. Barring new bugs, the app is a candidate for release to the wild.
- As an application moves further along in the lifecycle, it's opened to more people to try it out, test it, provide feedback, etc.
  - Prototype applications may only be shown or made available to key stakeholders.
  - Release candidate applications may be distributed to customers that sign up for early access.

Distribution

- Once the application has been stabilized, it's time to publish it. There are a number of different distribution options, depending on the platform.
- iOS
  - Ad-Hoc Deployment – Ad-hoc deployment is intended primarily for development and testing and allows you to deploy to a limited number of properly provisioned devices. When you deploy to a device via Xcode, it is known as ad-hoc deployment.
  - In-House Deployment – In-House deployment is meant for internal distribution of corporate applications that aren't available publicly via the App Store.
  - Apple App Store – Apple's App Store is a globally available online application repository, and it allows developers to market and distribute their apps online with very little effort.

Distribution

A digital certificate is a file that proves the authenticity of a device, server, or user. Under public key infrastructure (PKI), your certificate contains a public key that can be used to verify signatures generated by your private key.

- Apple Certificate Types:

Certificate type	Certificate name	Description
iOS Development	iPhone Developer: Team Member Name	Used to run an iOS, tvOS, or watchOS app on devices and use certain app services during development.
iOS Distribution	iPhone Distribution: Team Name	Used to distribute your iOS, tvOS, or watchOS app on designated devices for testing or to submit it to the store.
Mac Development	Mac Developer: Team Member Name	Used to enable certain app services during development and testing.
Mac App Distribution	3rd Party Mac Developer Application: Team Name	Used to sign a Mac app before submitting it to the Mac App Store.
Mac Installer Distribution	3rd Party Mac Developer Installer: Team Name	Used to sign and submit a Mac Installer Package, containing your signed app, to the Mac App Store.
Developer ID Application	Developer ID Application: Team Name	Used to sign a Mac app before distributing it outside the Mac App Store.
Developer ID Installer	Developer ID Installer: Team Name	Used to sign and distribute a Mac Installer Package, containing your signed app, outside the Mac App Store.

Distribution

- Android
  - All Android applications must be signed before being distributed. Developers sign their applications by using their own certificate protected by a private key. This certificate can provide a chain of authenticity that ties an application developer to the applications that developer has built and released.
  - It must be noted that while a development certificate for Android can be signed by a recognized certificate authority, most developers do not opt to utilize these services, and self-sign their certificates.
  - The main purpose for certificates is to differentiate between different developers and applications.
  - More details about Android application signing and publication to Google Play will be covered later in this course.
  - Unlike other popular mobile platforms, Android takes a very open approach to app distribution. Devices are not locked to a single, approved app store. Instead, anyone is free to create an app store, and most Android phones allow apps to be installed from these third-party stores.

Performance Tracking and Enhancement

- You can declare your mobile application a success or a failure by doing proper analysis:
  - The number of downloads
  - Total number of minutes spent by each user
  - Total revenue collected from the mobile app
- Remark: Mobile app release is just the first step in developing a continuous evolution, not the final product. Developers usually release the application with the minimum, and then listen to what the customer loves, hates, desires or even critical reviews. All these give them scope to improve!

Chapter 2.

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End

